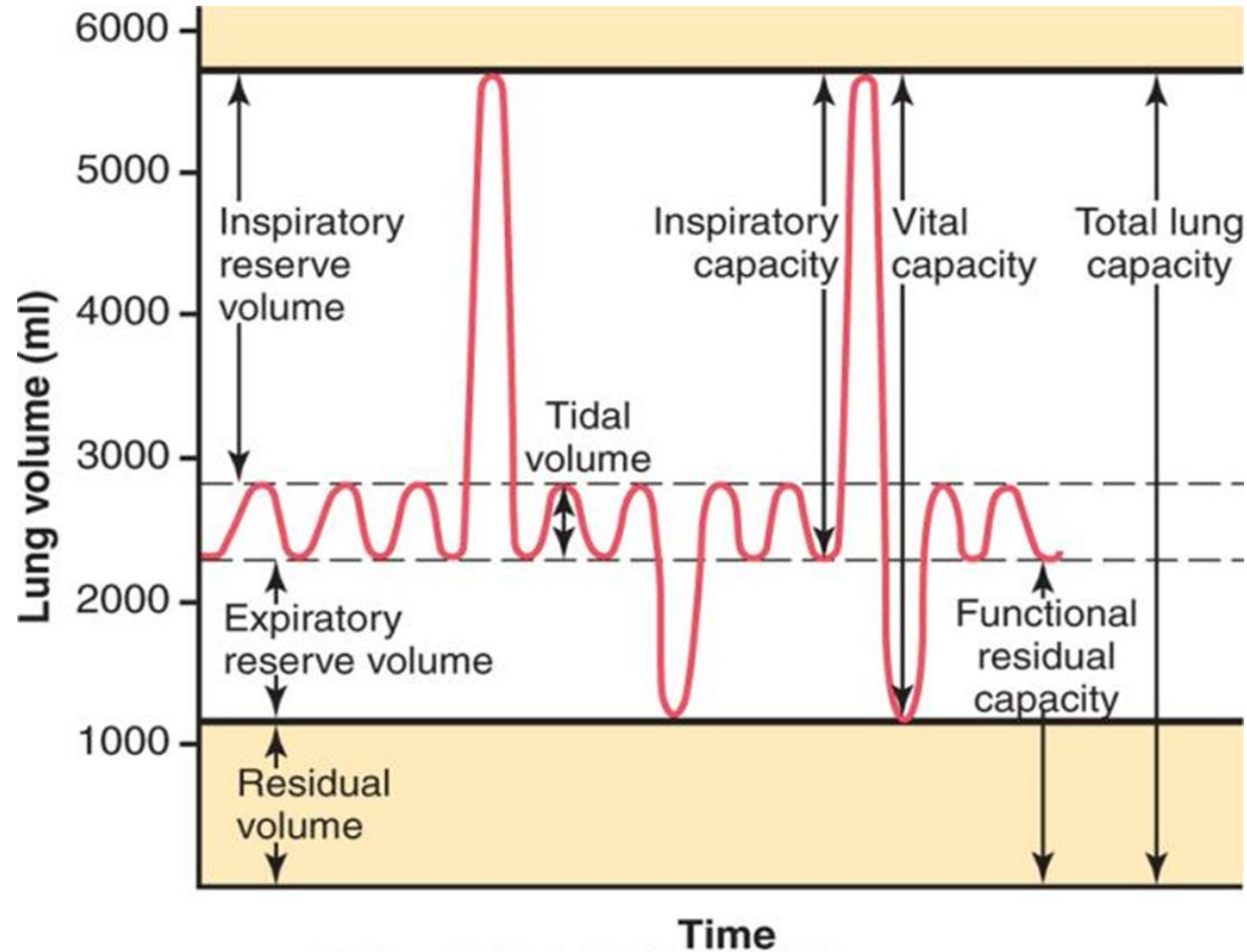


Learning Objectives

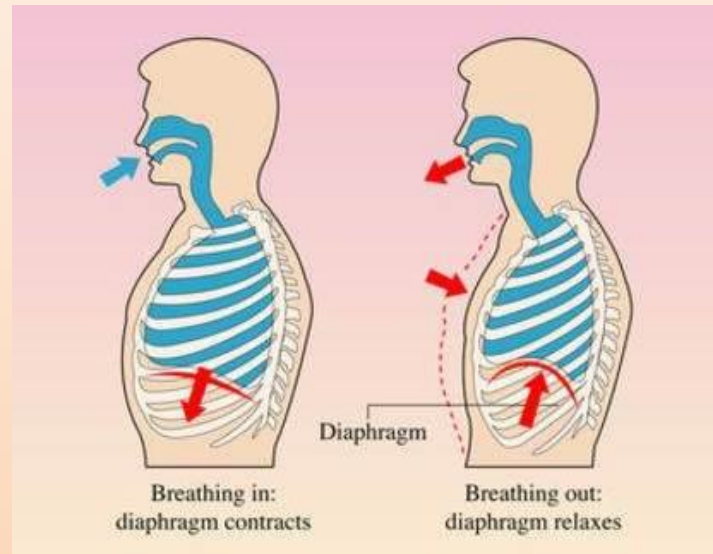
1. Distinguish between pulmonary & alveolar ventilation in normal respiratory physiology.
2. Apply muscle anatomy involved in the generation of pulmonary pressures.
3. Deconstruct lung mechanics generating intrapleural, alveolar and transpulmonary pressures.
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Lung Volumes and Capacities



Ventilation

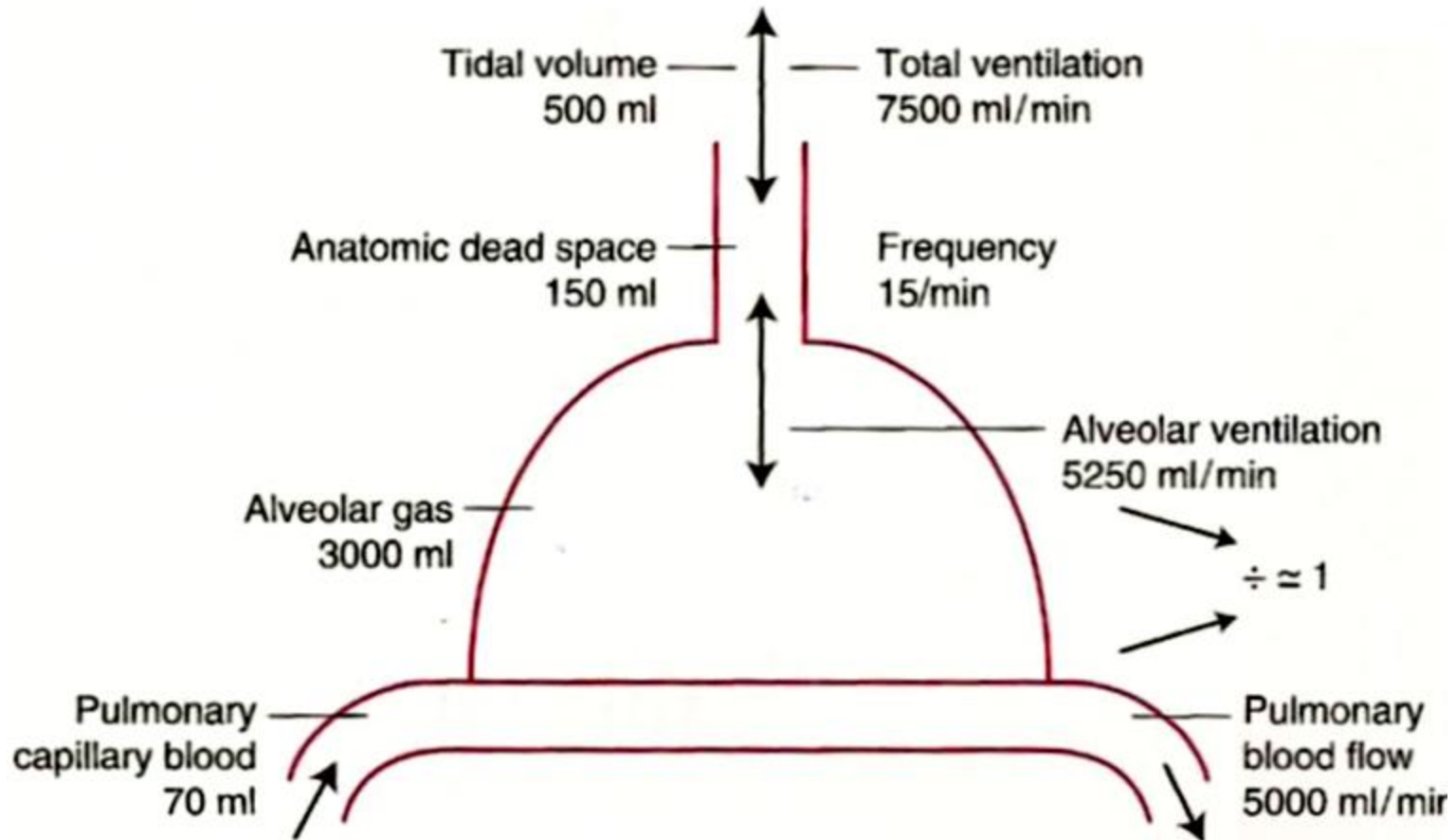
Ventilation occurs as a result of mechanical forces associated with the chest wall and lungs



Pulmonary and Alveolar Ventilation

- **Pulmonary Ventilation**
 - Gross ventilation per minute
 - **Tidal volume (V_T) x Respiratory Rate (RR)**
- **Alveolar Ventilation**
 - Fresh air reaching gas exchange areas of the lungs
 - **$(V_T - V_D) \times RR$**

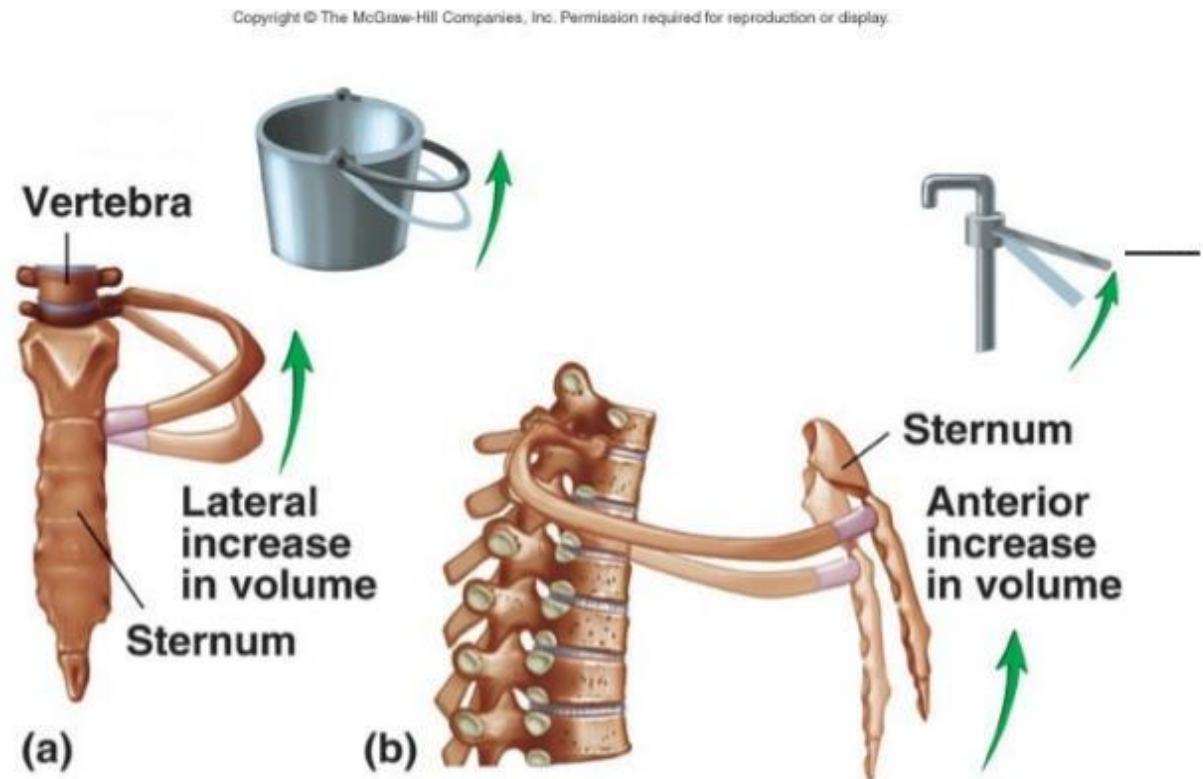
Summary - Ventilation



Muscles causing Lung Expansion and Contraction

External Intercostals:

- Upward/forward movement of rib cage
- BUCKET HANDLE MOTION



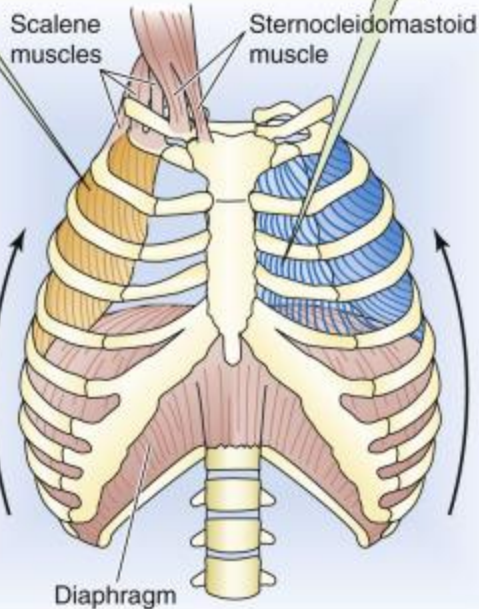
- In **resting breathing**, inspiration is active, but expiration is passive (or mainly passive).
 - Diaphragm
- In **vigorous breathing**, both inspiration and expiration are active
 - **Inspiration:** Diaphragm + accessory muscles of inspiration
 - **Expiration:** Accessory muscles of expiration



Summary

A INSPIRATION

The most rostral and dorsal subsets of the **external** intercostal muscles (gold)—as well as the parasternal subset of the **internal** intercostal muscles (blue)—have an *inspiratory* mechanical advantage.

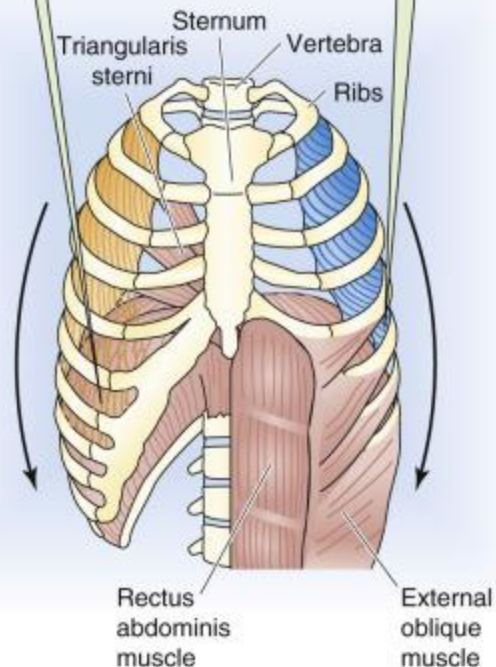


B BUCKET-HANDLE AND WATER PUMP-HANDLE EFFECTS

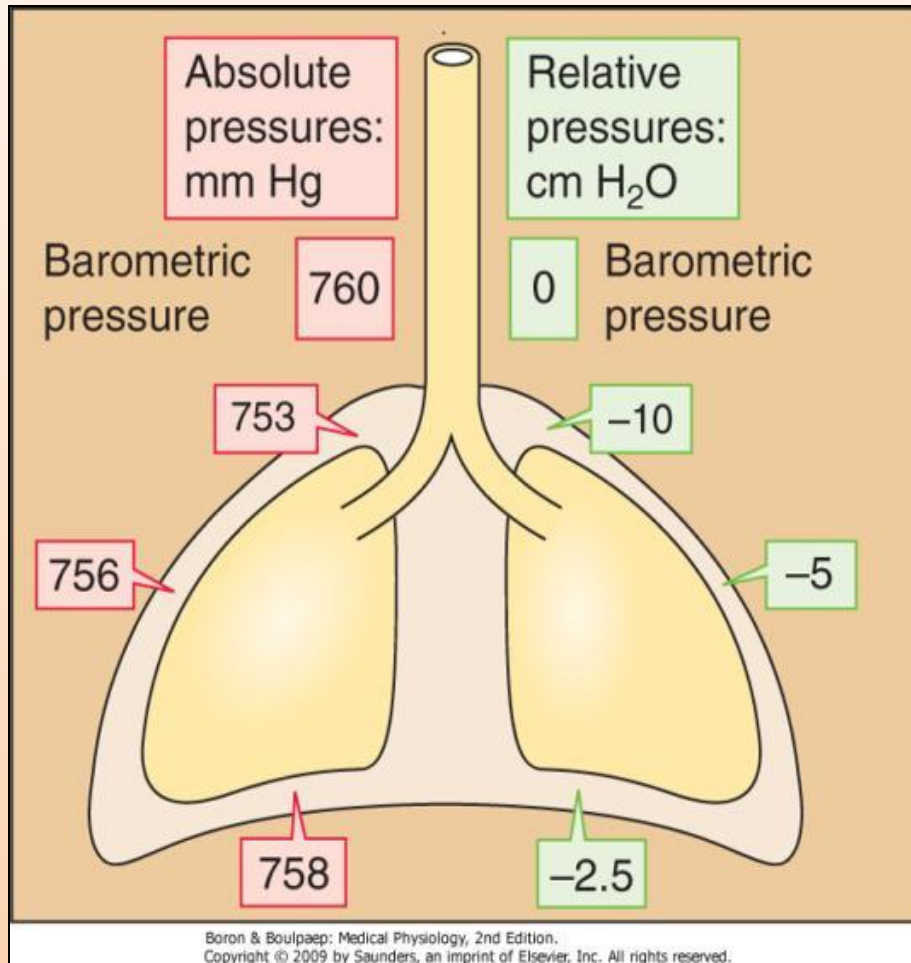


C EXPIRATION

The most caudal subset of the **internal** intercostal muscles (blue)—as well as the caudal-ventral subset of the **external** intercostal muscles (gold) and the triangularis sterni muscle (transversus thoracis)—have an *expiratory* mechanical advantage.



Pressures acting on the System

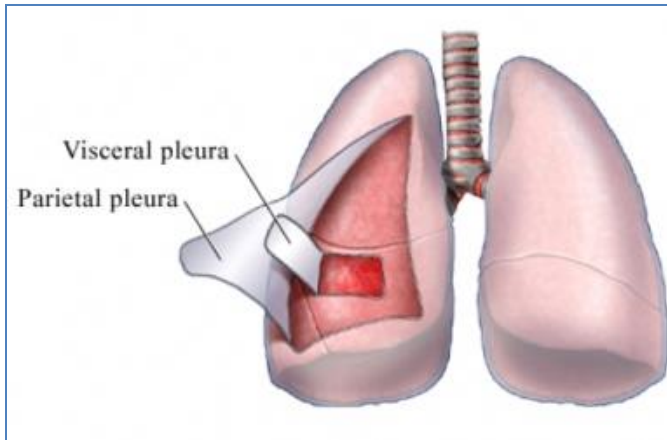


1. Intrapleural Pressure

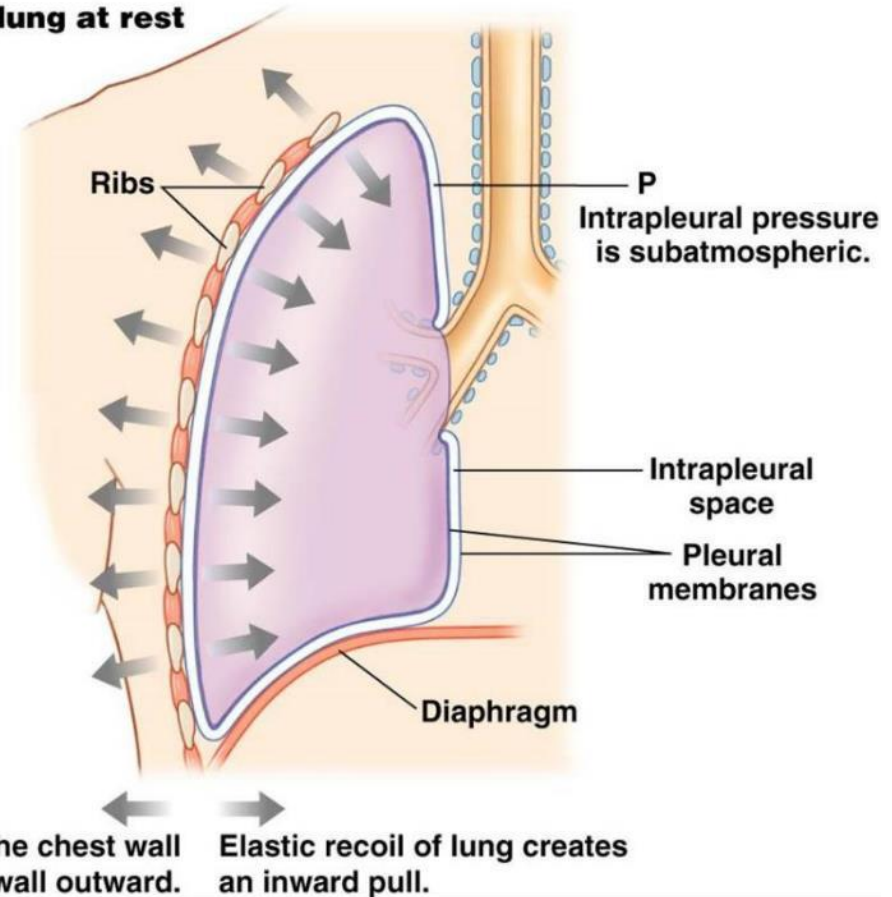
2. Alveolar Pressure

3. Transpulmonary Pressure

1. Intrapleural Pressure

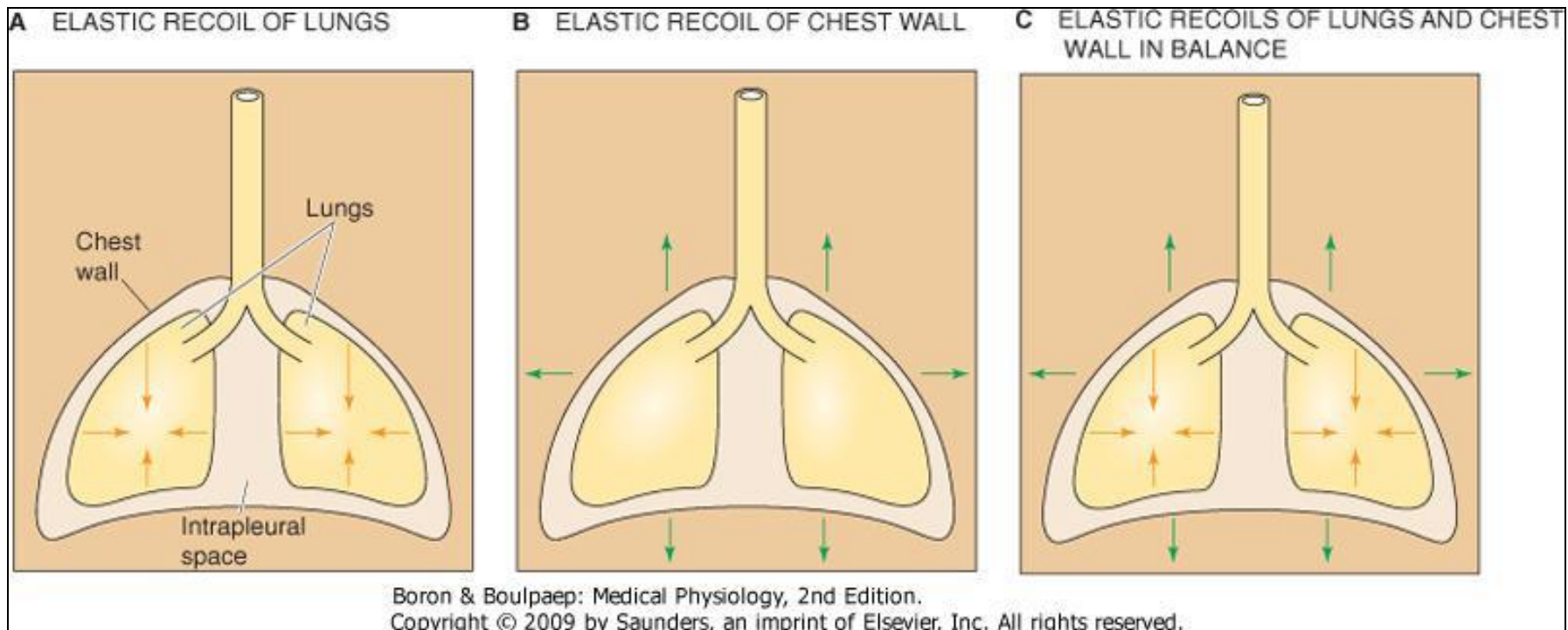


(a) Normal lung at rest

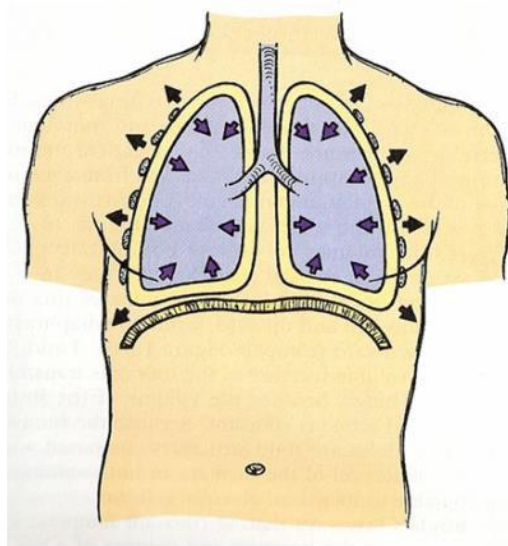


Negative Intrapleural Pressure

1. Lymphatic system drains the pleural fluid, generating a negative pressure
2. Elasticity of lungs and chest wall

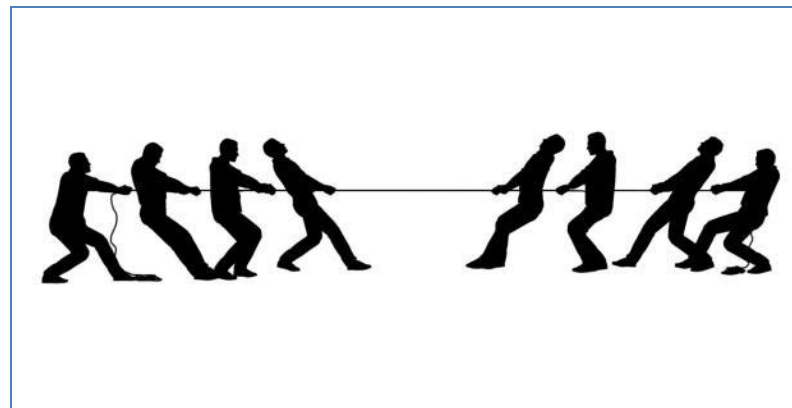


Both chest wall and lung have elastic recoil properties



Outward elastic recoil pressure of the chest wall

Inward elastic recoil pressure of the lungs



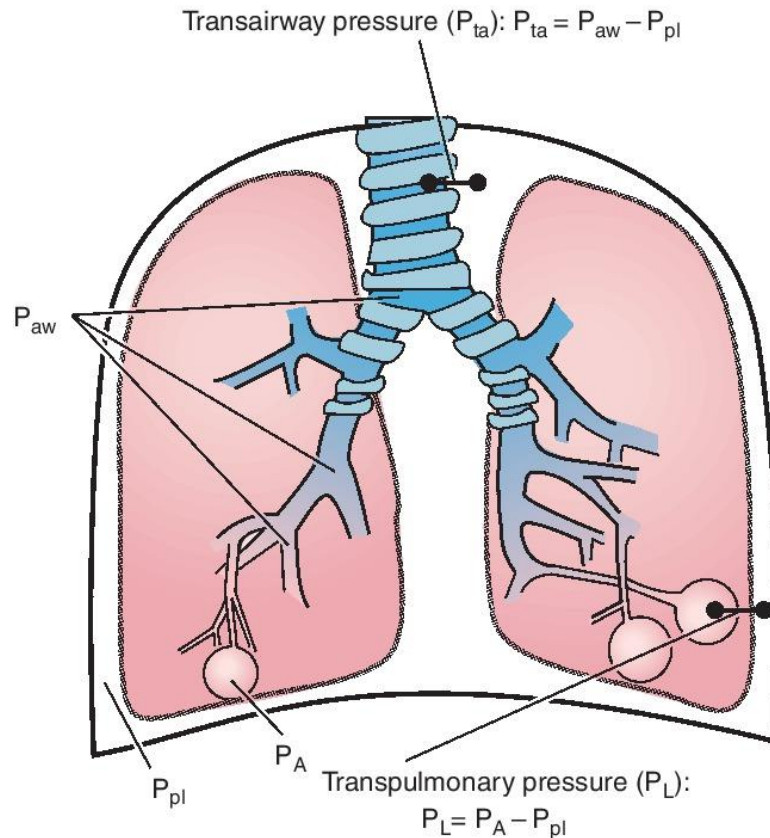
Lung Force Relationships During Inspiration

- Increase in negativity of intrapleural pressure due to contraction of the diaphragm and expansion of the chest wall
- Expansion of chest wall causes gas in the alveoli to expand creating negative alveolar pressure
- Increase in negative pressure in the lung (i.e. alveoli) causes air to flow in to the lungs

Lung Force Relationships During Expiration

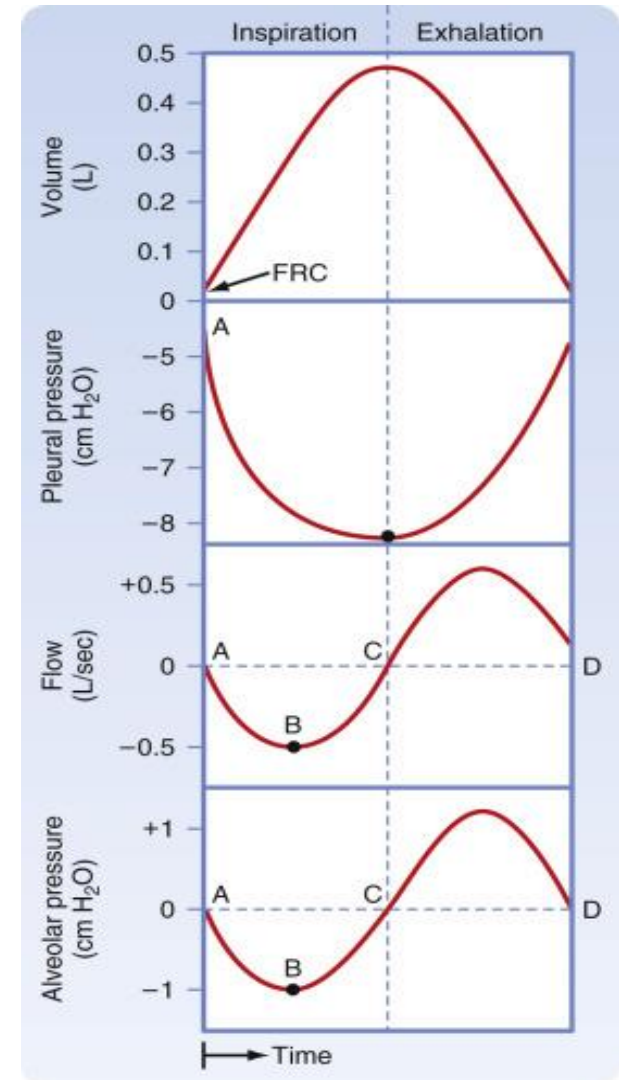
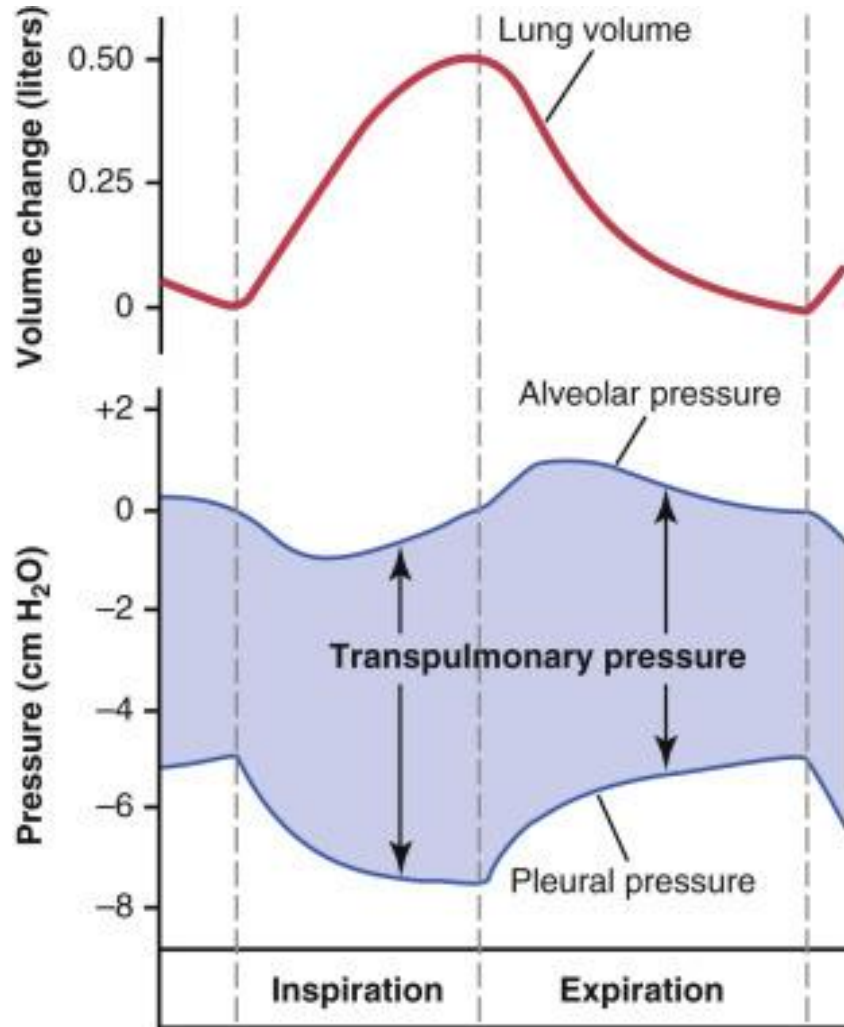
- Relaxation of the muscles of inspiration causes intrapleural pressure to return to -5 cm H₂O under resting conditions.
- Lung deflation begins and continues until the recoil force decreases to again equal intrapleural pressure. Once this occurs, the lung system is back to FRC.
- Deflation of the lung compresses the gases in the alveoli, creating a slightly positive alveolar pressure. This causes air to flow out of the lungs.
- The outflowing air returns alveolar pressure towards zero; at zero, airflow stops.

3. Transpulmonary pressure



- Several important pressures are involved in breathing: airway pressure (P_{aw}), **alveolar pressure (P_A)**, **pleural pressure (P_{pl})**, **transpulmonary pressure (P_L)**, and transairway pressure (P_{ta}). Both **transpulmonary pressure** and transairway pressure can be defined as the pressure inside minus the pressure outside. In both cases, the pressure outside is pleural pressure (P_{pl})
- **P_L** - Measure of the elastic forces in the lungs that tend to collapse the lungs at each instant of respiration, called the ***elastic recoil pressure***

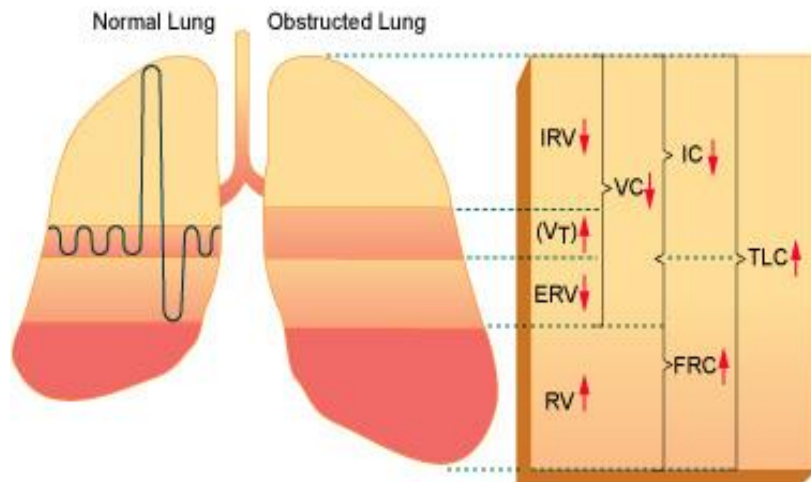
Pressures Acting on the System



Summary

- The outward elastic recoil of the chest wall (pleural pressure – pressure at surface of chest) is opposed by the inward elastic recoil of the lungs (alveolar pressure – intrapleural pressure).
- **Transpulmonary pressure** is (**elastic recoil pressure**) the **difference** between the **alveolar pressure** and the **intrapleural pressure** in the lungs. During ventilation, air flows because of pressure gradients.

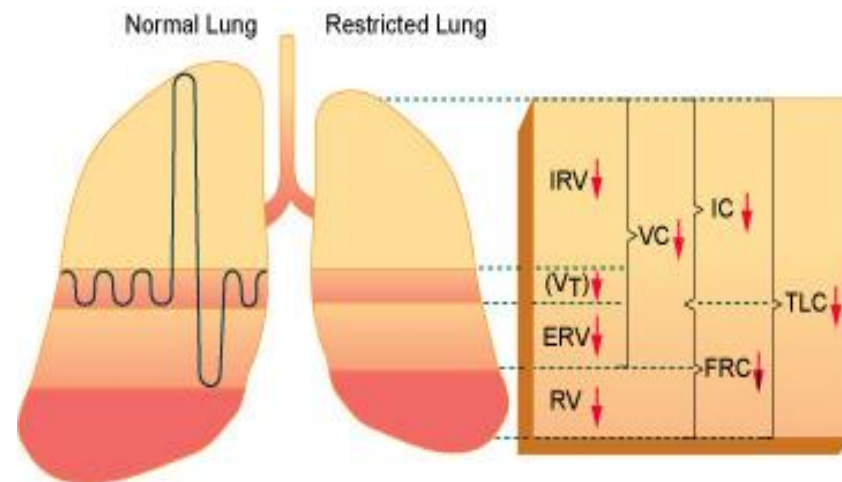
Obstructive Lung Disease



***Loss of ability to develop
+ alveolar pressure***

- Increased RV and FRC (both of these represent air left in the lung after expiration).

Restrictive Lung Disease



***Loss of ability to develop
- alveolar pressure***

- Decreased FRC (because less air is taken in with inspiration)
- Decreased TLC (because lungs are hard to expand and can't "stretch" as much on inspiration)

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